Program : **Diploma in Instrumentation Engineering**	
Course Code : **4082**	Course Title: **Microcontroller Programming and Applications**
Semester : **4**	Credits: **4**
Course Category: **Program Core**	
Periods per week: **4 (L:3 T:1 P:0)**	Periods per semester: **60**

Course Objectives:

- To develop background knowledge and core expertise of micro controller.
- To know the importance of different peripheral devices and their interfacing to micro controllers.
- To write assembly language programs of micro controllers for various applications.

Course Prerequisites:

Topic	Course code	Course name	Semester
Digital fundamentals		Digital Circuits and systems	3
Electronics fundamentals		Applied physics II	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Draw and describe architecture of 8051	14	Understanding
CO2	Explain the timers, interrupts, stack, serial data communication and power saving modes of 8051	15	Understanding
CO3	Write assembly language program for microcontrollers	15	Applying
CO4	Interface peripheral devices to the micro controllers.	14	Applying
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Draw and describe architecture of 8051**		
M1.01	Describe the Pin details of 8051	4	Understanding
M1.02	Describe the block diagram of micro controller 8051	5	Understanding
M1.03	Explain the internal ram organization	2	Understanding
M1.04	Describe special function registers in 8051	3	Understanding
Contents: Features of 8051- pin details of 8051-dual function of port3 pins- block diagram of micro controller 8051- internal ram organization - special function registers - PSW register T MOD-T CON-S CON-IE-IP-PROGRAM COUNTER-DATA POINTER-STACK POINTER – P CON - S BUFF-			
CO2	**Explain the timers, interrupts, stack, serial data communication and power saving modes of 8051**		
M2.01	Explain the timer modes of operation	4	Understanding
M2.02	Describe the interrupts	4	Understanding
M2.03	Describe the stack	1	Understanding
M2.04	Describe Serial data communication	4	Understanding
M2.05	Explain the power saving modes of 8051	2	Understanding
	Series Test – I	1	

Contents:

Timer modes of operation –mode 0, mode1, mode2, mode3-counting-machine cycle-Stack-Interrupts-Types--Priority- Serial data transmission –data reception- Power saving mode and idle mode

CO3	**Write assembly language program for micro controller**		
M3.01	Describe the Addressing modes of 8051	2	Understanding
M3.02	Explain the Instruction set of 8051	4	Understanding
M3.03	Write assembly language programs	9	Applying

Contents:

Addressing modes- Instruction set- arithmetic, logic, rotate, data transfer, and branch instructions-loop, jump and call instructions-assembly language program such as largest of an array, array sorting, code conversion HEXA to ASCII, 2's complement, square of a number, occurrence of a byte, multiplication, division, Multibyte addition

CO4	**Interface peripheral devices to the micro controller**		
M4.01	Interface LCD	3	Applying
M4.02	Interface key board(4x4)	2	Applying
M4.03	Interface ADC and DAC	3	Applying
M4.04	Interface temperature sensor LM 35	2	Applying
M4.05	Drive relay using 8051	1	Applying
M4.06	Interface an Opto isolator	1	Applying
M4.07	Interface DC motor and Stepper motor	2	Applying
	Series Test – II	1	

Contents:

Interfacing- LCD (16x2) LCD commands, keyboard(4x4),ADC and its pin details, DAC and its pin details, temperature sensor LM35, Relays Opto isolators, DC motor and Stepper motor

Text / Reference:

T/R	Book Title/Author
R1	Kenneth J Ayala - Thomson ,The 8051 Microcontroller , Third Edition Pearson
R2	Subratha Ghoshal , 8051 microcontroller internals, instructions, programming and interfacing ,Pearson
R3	Muhammad Ali Mazidi and Janice Gillispie Mazidi , The 8051 Microcontroller and Embedded Systems Using Assembly and C - Second Edition ,Pearson Education
R4	Muhammad Ali Mazidi, Sarmad Naimi and Sepehr Naimi -The AVR Microcontroller and Embedded Systems using assembly and C - - Pearson Education.
R5	Dr. K V K Prasad- Embedded / Real Time Systems Concept, Design and Programming The ultimate reference - (Dreamtech).

Online Resources:

Sl.No	Website Link
1	https://www.electronicshub.org/8051-microcontroller-introduction/
2	https://www.tutorialspoint.com/embedded_systems/es_microcontroller.htm
3	https://www.electronicwings.com/8051/introduction-to-8051-controller
4	https://nptel.ac.in/courses/Webcourse-contents/IIT-KANPUR/microcontrollers